DAY 10 | HANDS-ON ACTIVITIES

React Basics: Problem Statements

Seven problems. In Colab, use show_react() from the notebook. In VS Code, use a Vite React project and edit src/App.jsx.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Hello Component

Level: Warm-up | **Concepts:** component, JSX expressions

Create a Greeting component that shows your name, today's date and a message that says "Good morning" or "Good evening" based on the current hour.

Sample run / expected result

```
Hello, Aiko!
Today is Fri Nov 06 2026
Good morning
```

Hints

- new Date().getHours() < 12

Problem 2: Ticket Cards

Level: Easy | **Concepts:** props, reuse

Create TicketCard with props id, title and priority. Render three cards for three different SmartDesk tickets. High priority cards show a red border.

Sample run / expected result

```
[T-101 | No signal at home | high]
[T-102 | Double charge | medium]
[T-103 | Change plan | low]
```

Hints

- style={{ borderLeft: priority === "high" ? "4px solid #C2410C" : "none" }}

Problem 3: Counter with Limits

Level: Easy | **Concepts:** useState, onClick

Build a counter with +1, -1 and Reset buttons. The count must never go below 0 or above 10; disable buttons at the limits.

Sample run / expected result

```
Count: 10 [+1 disabled] [-1] [Reset]
```

Hints

- disabled={count === 10}

Problem 4: Live Converter

Level: Medium | **Concepts:** controlled input, derived values

Build a Celsius input that shows Fahrenheit and Kelvin as you type. Show "Enter a number" when the input is empty or not a number.

Sample run / expected result

```
Celsius: 37
Fahrenheit: 98.6 | Kelvin: 310.15
```

Hints

- Number(value) and Number.isNaN()

Problem 5: Layout with children

Level: Medium | **Concepts:** children prop, composition

Create a Panel component with a title prop and children. Build a page with three panels: About, Skills and Contact, each with different content.

Sample run / expected result

```
Three styled panels, each with a heading and custom content
```

Hints

- function Panel({ title, children })

Problem 6: Review the Codex Dashboard

Level: Challenging | **Concepts:** code review, keys, state, secrets

Ask Codex to generate the SmartDesk dashboard (App, Header, FilterBar, TicketList, TicketCard). Review it with the checklist from the deck: keys, no mutation, no secrets, no raw HTML, small components. Write a review note with at least two issues you found or confirmed absent, and fix them.

Sample run / expected result

```
Review note
1. TicketList used the array index as key -> changed to ticket.id
2. Filter mutated the tickets array with sort() -> now sorts a copy
3. No secrets found
```

Hints

- Ask Codex: "List every place this code uses state, and how it changes it."

Problem 7: Mini Signup Form

Level: Stretch | **Concepts:** multiple controlled inputs, onSubmit, validation

Build a form with name, email and password. Show validation messages as the user types (email must contain @, password at least 8 characters). On submit, show a welcome message instead of the form.

Sample run / expected result

```
Password must be at least 8 characters
... after submit: Welcome, Mei!
```

Hints

- Use one state object: const [form, setForm] = useState({ name: "", email: "", password: "" })
- e.preventDefault() in onSubmit